



University  
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# OKD4 Cluster

Introduction

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# What is the **problem** we are trying to solve?

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- The reality is that much of the research that we do within the IDA section **cannot be done on a local laptop or desktop**.
  - Deep learning or AI research that needs powerful GPUs for training and inference
  - Big document collections that require 100's GB of memory to load
  - Computationally intensive jobs that need to be distributed and processed in parallel, so you are not waiting for weeks to get a result
  - You require services that need to run continuously (i.e. don't shutdown when you close your laptop lid)
  - .... and many more

Our compute cluster can solve these issues!

# What is the Cluster?

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- The IDA compute cluster is a **managed** set of **hardware** and **software** that is designed to provide **on-demand access to compute capacity** (both CPU and GPU based).
- For reference we will refer to it as the **OKD4** Cluster (after the management platform it uses)

## **OKD4** Provides:

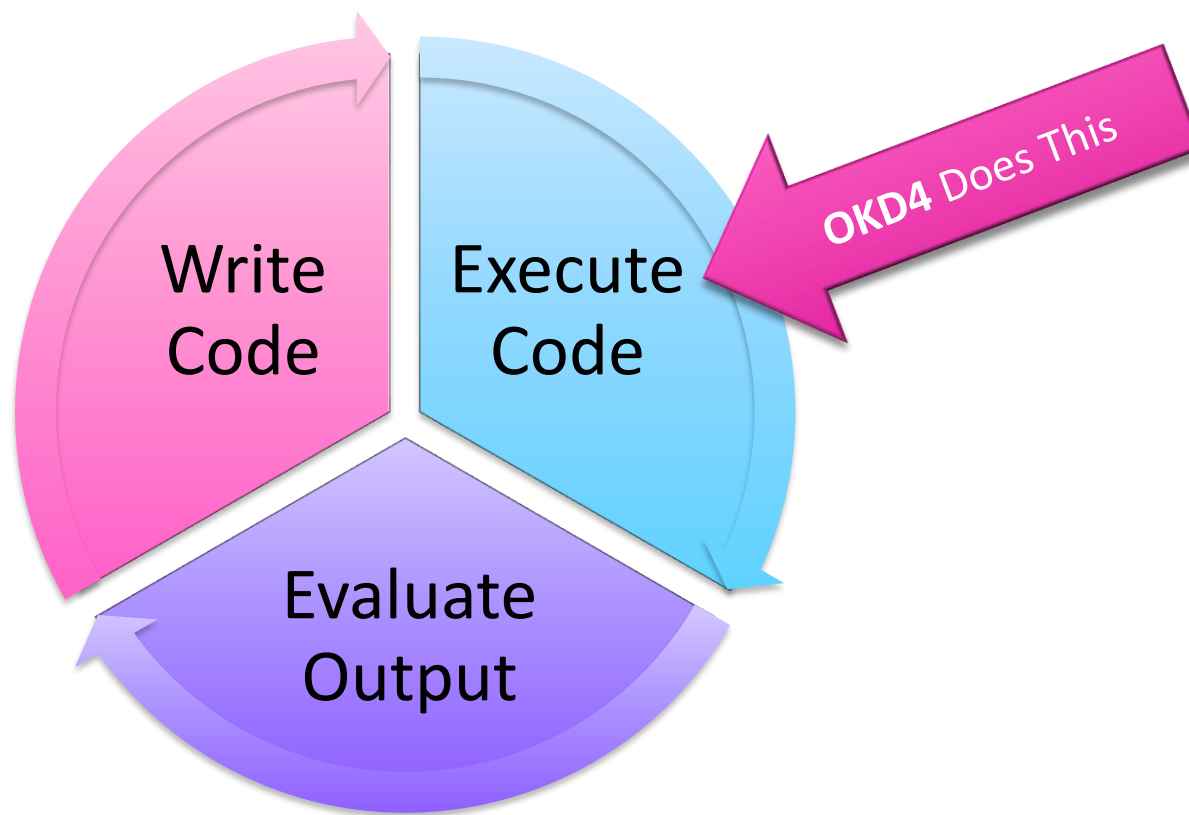
- ✓ On-demand container-based sandboxes for your research
- ✓ Fast setup for common use-cases (like Jupyter notebooks) via the Launcher service
- ✓ A range of GPU types
- ✓ Persistent data storage via NFS and Ceph
- ✓ Shared data directories for common datasets, indices, LLMs and more
- ✓ Continuous integration for application developers

.... and much more

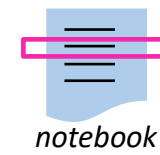


# Conceptually, what does it do?

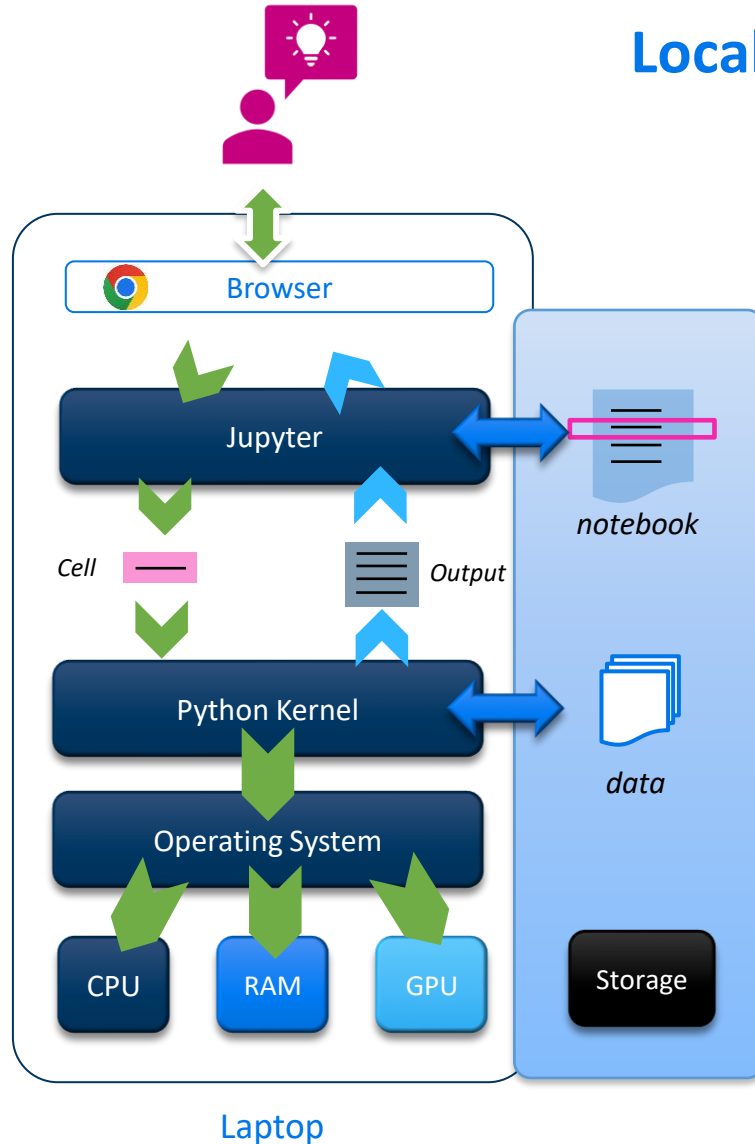
- Most of the work that we do can be divided into three phases: writing code, executing that code, and evaluating the output – OKD4 allows you to **offload** the **execution** part to a (powerful) remote machine (or multiple machines)



# Use-Cases: Python Notebooks

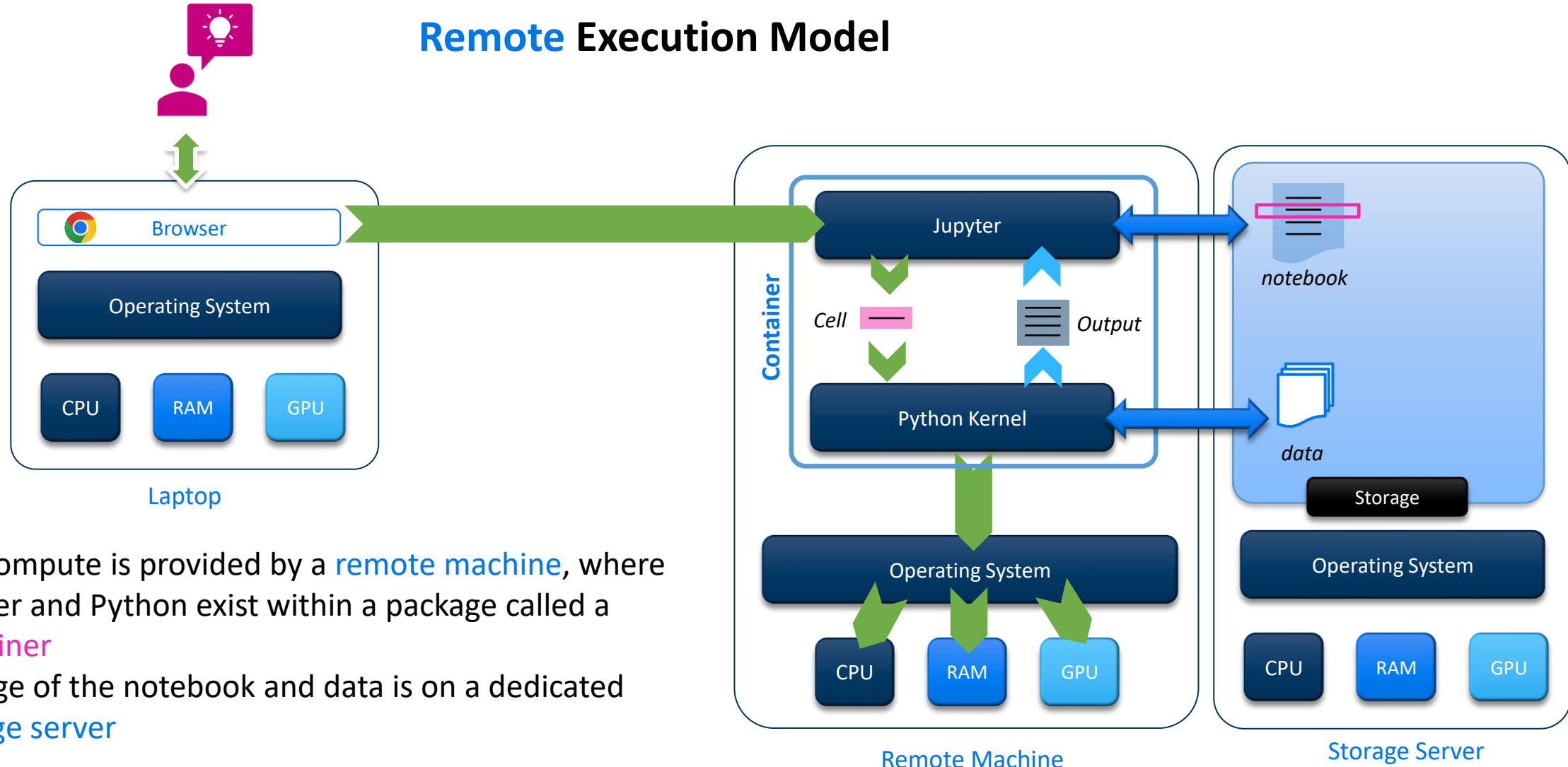
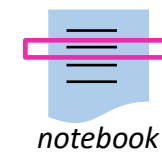


## Local Execution Model



- All compute happens on the local machine
- Reliant on local resources (CPU, RAM, GPU and Storage)
- Data is written to local storage

# Use-Cases: Python Notebooks



# Why is Remote Execution Better?

## Higher Capacity

- Compute capacity is not capped by your local machine, which must make trade-offs that a server does not need to (weight, power, noise, etc.)

## Resource Pooling

- No-one uses their computer 24-7, so there is time when that machine is idle, in a remote execution setting we can simply serve idle capacity to someone else who needs it

## Scalability

- You can request what capacity that you need at the time, allowing for you to scale up for complex tasks, or distribute your jobs across multiple machines

## Background Processing

- The cluster runs 24-7, and so can still be processing your work when your laptop is off

## Remote Storage

- If your laptop gets broken, your data is still stored on the remote storage server

## Shared Resources

- We can share commonly shared resources (e.g. datasets or LLMs) more easily via Storage Server

# How Do I Use it?

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- You will first need an **OKD4 user account**, these are available to:
  - Any staff and post-doctoral researchers within the IDA Section
  - Any PhD students who's primary supervisor is within the IDA Section
  - Level 4, MSci and MSc project students who have a supervisor within the IDA Section
- You can request an account by e-mailing, the cluster administrator, currently:
  - [richard.mccreadie@glasgow.ac.uk](mailto:richard.mccreadie@glasgow.ac.uk)
- If you are a student, you should cc your supervisor in the email and remember to state what type of student you are.



# How Do I Use it?

What you **need to do** will define **what you need to learn** to use the cluster, you can think of it as having **4 levels**, where you incrementally need to understand more of how the cluster works as you need to undertake more complex work

## Level 0: Understanding your account, resources and storage



### Level 1

- Cluster users who just need to get up and running with Python/Jupyter or need remote execution with VSCode are recommended to use the Launcher tool



### Level 2

- For scenarios where you need to deploy a wider range of existing services, you will need to learn the basics of Containers and Kubernetes. This will allow you to deploy pre-built container solutions from platforms like Dockerhub or more complex services via OperatorHub.



### Level 3

- For running batch experiments (queuing many containers to run concurrently) you will need to learn how to interact with the cluster via the command line tool (oc)



### Level 4

- If you need to create a new development environment from first principles or custom applications/demos, then you will need to learn how to produce new containers via Builds, and it is recommended to learn how to create continuous integration pipelines.

# Questions?

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